VINAY K R

Software Engineer | Backend & Full-Stack | Applied AI

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SUMMARY

Software engineer with full-stack experience at Light & Wonder and Liminal Custody, plus co-founder engineering work at Medha Innovations. Builds backend services and applied-AI workflows using Go, TypeScript and Python/FastAPI, with emphasis on transactional correctness, authorization, evaluation, reliability and human review.

PROFESSIONAL EXPERIENCE

Medha Innovations

Founding Engineer / Co-founder | Mar 2026 - Sep 2026

- Co-founded a self-funded booking and messaging venture; designed a Go modular backend across 21 domains, with REST APIs for profiles, practitioner discovery, availability and bookings.
- Designed booking coordination around PostgreSQL transactions and row locking, with PostGIS-backed location discovery and explicit data-model/API boundaries.
- Platform design combined RS256 JWT/RBAC with Redis Pub/Sub-backed WebSockets. Collaborated with a co-founder on integration and deployment planning; project configuration includes Docker and Kubernetes/Argo CD.

Liminal Custody (First Answer India Services Pvt. Ltd.)

Software Engineer, Full Stack | Nov 2025 - Mar 2026

- Built a transaction-policy engine for three controls (TRM Labs address-risk screening, FATF Travel Rule checks, transfer velocity limits) as TypeScript APIs with Redis-cached, short-circuit rule evaluation.
- Implemented tenant-level RBAC, quorum approvals, Auth0 authentication and step-up two-factor authentication (2FA) with signature validation, plus TypeORM transactions that prevent partial updates.

Light & Wonder

Senior Associate Software Engineer (final role) | Aug 2023 - Jul 2025

- Built an internal LLM-assisted debugging tool (Gemini API, React UI), deployed on Microsoft Azure with the DevOps team and used by internal teams, that matched crash reports and telemetry to similar historical hardware incidents and their fixes.
- Selected for the slot-machine modernization team; built an in-game LLM chat assistant (OpenAI API, Angular) that lets guests order food and beverages and choose music from their seat; demoed to management and approved for lab testing.
- Fixed a memory leak that made slot machines hang or restart after about one hour: reproduced it on hardware, corrected component cleanup and resource reuse, and verified sustained operation.
- Built SQL Server-backed APIs and screens for casino cash operations; automated audit capture with triggers and stored procedures; wrote Cypress end-to-end tests.
- Worked on CONNEXUS, the slot-machine tracking and data management system that monitors machine status and usage and collects data on a central server; contributed to performance profiling and monitoring.

Light & Wonder

Intern | Mar 2023 - Jul 2023

TECHNICAL SKILLS

Languages and Frameworks: Python, TypeScript, Go, SQL; FastAPI, Pydantic, asyncio, Node.js, REST APIs, Server-Sent Events (SSE), WebSockets, React and Angular.

LLM and Workflows: Gemini, Groq and OpenAI-compatible APIs; prompt design, structured outputs, JSON Schema, tool calling, LangGraph, Model Context Protocol (MCP), checkpointed workflows, content-bound approvals and bounded execution.

Retrieval and Search: PostgreSQL, SQLite FTS5, BM25, embeddings, hybrid retrieval, reciprocal rank fusion, ONNX Runtime, cross-encoder reranking and quantization.

Document AI: Docling, OCR, vision-language models, Pydantic-validated extraction, evidence bounding boxes, decimal-exact reconciliation and human-review routing.

Testing and Observability: pytest, Go race tests, held-out and synthetic evaluations, F1, Recall@k, latency/cost tracking, OpenTelemetry, Prometheus, Grafana and Alertmanager.

Data and Delivery: PostgreSQL, PostGIS, Redis, SQLite, SQL Server, Docker, GitHub Actions and GitLab CI/CD; Kubernetes (k3s) and Argo CD deployment configuration.

PROJECTS

TraceWard: Incident-Triage Workflow

https://github.com/vi-nayKR/fastapi-genai-agent-patterns

- Built a checkpointed, predefined LangGraph workflow for log parsing, similar-incident retrieval, root-cause suggestions and fix drafting, using MCP tools over stdio and OpenTelemetry tracing.
- Persists a local ticket draft only after authenticated, content-bound approval; enforces tenant scope, signed approval, idempotency and execution budgets. Does not file external issues or execute fixes.
- Evaluated 50 synthetic known-family holdouts against a 200-incident corpus: 50/50 root-cause top-1; 20 approval-bypass probes produced no unauthorized drafts. Not general incident accuracy or security.

ReconcileAI: Document AI Reconciliation

https://github.com/vi-nayKR/multimodal-document-intelligence

- Built a local invoice/PO workflow combining Docling parsing, Pydantic-validated model extraction and decimal-exact discrepancy checks, with source-evidence links and human review.
- Compared extraction on 20 held-out public CORD receipts: Gemini vision reached 76.0% field F1 versus 66.0% for a Qwen VLM and 56.9% for OCR plus text LLM; not invoice/PO accuracy.
- Verified deterministic controls on 50 labelled synthetic invoice/PO pairs; separately measured live model behavior, including review burden and row-scoring limitations. Streams progress over SSE.

Homelab SRE: Observability

https://github.com/vi-nayKR/homelab-sre-observability/tree/hardening/production-readiness

- Deployed an isolated six-service observability stack on a shared Linux host, with loopback-only access, file-backed secrets, authenticated alert reception and resource limits. CI covers Go race tests, alert lifecycle, restart recovery and cold-backup/file-restore checks. Single-host scope, not HA or independent paging.

EDUCATION

Siddaganga Institute of Technology (SIT), Tumakuru | 2019 - 2023

B.E., Computer Science and Engineering | CGPA: 8.65/10

PUBLICATIONS AND CERTIFICATIONS

Data Visualisation of Time Tradable Assets Using Machine Learning | IEEE NMITCON 2023

https://doi.org/10.1109/NMITCON58196.2023.10275962

Model Context Protocol: Introduction and Advanced Topics | Anthropic

Docker Foundations Professional Certificate | LinkedIn Learning